

LESSON 02: EFFECT OF CLIMATE ON CROP CULTIVATION

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Course: Agricultural Technology / Crop Science

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2.1 Introduction to Climatic Parameters Affecting Crop Cultivation & Their Measurement

Weather

Weather refers to the short-term state/condition of the atmosphere at a specific time and location.

Climate

Climate refers to the long-term average pattern of weather conditions prevailing over a particular region.

Climatic Factors

- Rainfall / Precipitation
- Relative Humidity
- Temperature
- Light / Solar Radiation
- Wind

Rainfall (Precipitation)

Rainfall is defined as water falling from clouds to the ground in the form of droplets with a diameter ranging between **1 mm to 5 mm**.

Measurement of Rainfall

Rainfall is measured as the depth/height of water that accumulates on a flat surface, assuming no runoff, absorption, or evaporation occurs. It is measured using non-recording rain gauges and automatic rain gauges.

Non-Recording Rain Gauge (Standard/Simple Rain Gauge)

4 Main Components:

1. Funnel
2. Outer Cylindrical Container
3. Inner Water Collector Container
4. Measuring Cylinder (specially calibrated)

Key Details & Specifications:

- **Funnel Diameter:** *12.7 cm*
- **Sharp Edge:** The upper rim of the funnel features a very sharp edge to prevent raindrops from splashing outward upon impact.
- **Observation Time:** Readings are recorded at **9:00 AM** daily at the meteorological station.
- **Measurement:** Collected water in the inner container is poured into a calibrated measuring cylinder and read as height in **millimeters (mm)**.

Sample Calculation Problem:

Problem Statement: Express the rainfall received in a region as a height (depth in mm) if the volume of water collected in a simple rain gauge on a given day is *10 cm³*.

Solution Steps:

- Funnel Diameter (*d*) = *12.7 cm*
- Funnel Radius (*r*) = $12.7 / 2 = 6.35 \text{ cm}$
- Cross-sectional Area of Funnel (*A*) = $\pi r^2 = (22 / 7) \times (6.35)^2 \text{ cm}^2$
- Volume of collected water (*V*) = *10 cm³*
- Formula: **Volume (*V*) = Area (*A*) × Height (*h*)**
- $h = V / (\pi r^2) = 10 / [(22 / 7) \times 6.35 \times 6.35] \approx 0.079 \text{ cm} = 0.79 \text{ mm}$

Calculated Rainfall: *0.79 mm*

Automatic Rain Gauge

Automatically records the accumulated water volume over a 24-hour period onto a chart graph. Advantages include measuring:

- Total rainfall amount
- Duration of the rainfall event
- Rainfall intensity

| Light / Solar Radiation

Solar radiation reaches the Earth across various wavelengths, categorized into three main types:

1. Ultraviolet (UV)
2. Visible Light
3. Infrared (IR)

Visible light consists of 7 colors and affects crop growth through 3 key properties:

- **Light Intensity**

- **Light Quality (Wavelength composition)**
- **Photoperiod (Light Duration)**

Instruments for Light Measurement:

- **Campbell-Stokes Sunshine Recorder:** Used to measure the total daily sunshine hours.
- **Pyranometer / Actinometer:** Used to measure total solar radiation intensity. Solar rays fall on a bimetallic strip, causing differential thermal expansion that mechanically moves a pen arm across a clockwork chart cylinder.

Wind

Wind is the movement of air from one region of atmospheric pressure to another.

- **Anemometer:** Measures wind speed (recorded in **km/h**).
- **Wind Vane:** Indicates wind direction. The arrow head points directly into the direction from which the wind is blowing (e.g., in a wind blowing toward the Northeast, the arrowhead points toward the Southwest origin).

Temperature

Atmospheric temperature is measured using thermometers positioned at a standard height of **1.2 meters** above ground level inside a Stevenson Screen.

Maximum Thermometer

Measures the highest temperature reached during a given period. Features a narrow constriction near the bulb in the mercury column. As temperature rises, mercury expands past the constriction; as it cools, the mercury column breaks at the constriction, leaving the maximum level indicated.

Minimum Thermometer

Measures the lowest temperature during a given period. Uses alcohol as the liquid medium containing a lightweight dumbbell float (index). As temperature drops, surface tension pulls the float toward the bulb; as temperature rises, alcohol flows past the float, leaving it at the lowest point reached.

Six's Maximum-Minimum Thermometer: A U-shaped instrument capable of simultaneously measuring both maximum and minimum temperatures within a single unit.

Humidity

Humidity is the amount of water vapor present in the atmosphere.

Relative Humidity (RH)

Relative Humidity is the percentage ratio of the actual amount of water vapor present in a given volume of air at a specific temperature and pressure, compared to the maximum amount of water vapor required to saturate that same volume of air at the same temperature and pressure.

Measurement of RH (Dry & Wet Bulb Psychrometer):

Readings are taken from a Dry Bulb and a Wet Bulb thermometer. The temperature difference (depression) is calculated, and RH (%) is determined using standard Psychrometric Tables.

Example:

- Dry Bulb Reading = 30°C
- Wet Bulb Reading = 28°C
- Temperature Difference = $30 - 28 = 2^{\circ}\text{C}$
- Relative Humidity (from standard table) = **85%**

Stevenson Screen

A wooden shelter with louvered side panels (~5 cm wooden slats) painted white to minimize direct solar radiation absorption.

- **Purpose:** Protects delicate meteorological instruments from direct weather exposure while ensuring free natural air circulation.
- **Instruments Inside:** Maximum & Minimum Thermometers, Dry & Wet Bulb Thermometers, Hair Hygrometer.
- **Key Features:** Positioned **1.2 m to 1.5 m** above ground level to align with eye level for accurate reading and avoid ground reflectance heat. Doors face North/South (South in Southern Hemisphere/tropics context) to block direct sun entry when opening.

Evaporation

Evaporation is the process by which liquid water transitions into vapor from open water surfaces.

- **Class A Evaporation Pan:** Standard instrument made of galvanized iron, measuring **120.7 cm in diameter** and **25.4 cm in depth**. Painted white to minimize radiation heat absorption.
- **Installation:** Mounted on a wooden frame **15 cm** above ground. Water level maintained 5 to 7.5 cm from the top rim.
- **Measurement:** Calculated using a **Stilling Well** and a **Hook Gauge**.

Summary Table: Climatic Parameters, Instruments & Units

Climatic Parameter	Sub-Parameter	Measuring Instrument	Units
Light / Solar Radiation	Sunshine Duration	Campbell-Stokes Sunshine Recorder	Hours (h)
	Light Intensity	Pyranometer / Actinometer	W/m^2
Wind	Wind Speed	Anemometer	km/h
	Wind Direction	Wind Vane	Compass Directions
Temperature	Max / Min / Air Temp	Max, Min, or Six's Thermometer	$^{\circ}C$ / $^{\circ}F$
Relative Humidity	Atmospheric RH	Psychrometer (Dry & Wet Bulb) / Hair Hygrometer	%
Atmospheric Pressure	Air Pressure	Barometer	Millibar (mb) / Pascal (Pa)
Evaporation	Open Water Evaporation	Class A Evaporation Pan (with Hook Gauge)	Millimeters (mm)
Rainfall	Precipitation Depth	Simple / Automatic Rain Gauge	Millimeters (mm)

2.2 Effect of Climatic Factors on Crop Cultivation

Precipitation & The Hydrological Cycle

Precipitation refers to water falling from the atmosphere to the Earth's surface in liquid or solid form. The continuous movement of water through Earth's spheres is known as the **Hydrological Cycle**.

Rainfall Mechanism & Atmospheric Circulation

High-pressure belts exist at $30^{\circ}N$ and $30^{\circ}S$ latitudes, with a low-pressure belt around the Equator. Air masses flow constantly from high-pressure zones toward the equatorial low-pressure zone. In the Northern Hemisphere, winds blow from the Northeast; in the Southern Hemisphere, winds blow from the Southeast and veer Southwest upon crossing the Equator. The convergence zone near the Equator is called the **Intertropical Convergence Zone (ITCZ)**.

Rainfall Mechanisms Operating in Sri Lanka

1. **Monsoonal Winds:** Southwest Monsoon and Northeast Monsoon.
2. **Convictional Rain:** Thermal atmospheric instability leading to local afternoon thunder showers.
3. **Depressional / Cyclonic Systems:** Atmospheric low-pressure systems originating in the Bay of Bengal.

Rainfall Seasons & Rhythms in Sri Lanka

- **First Inter-Monsoon (FIM):** March – April (Countrywide convectional rains)

- **Southwest Monsoon (SWM):** May – September (Rains concentrated in the Wet Zone)
- **Second Inter-Monsoon (SIM):** October – November (Widespread convectional rains countrywide)
- **Northeast Monsoon (NEM):** December – February (Rains concentrated in the Dry and Intermediate Zones)

Feature	Southwest Monsoon (SWM)	Northeast Monsoon (NEM)
Seasonal Trigger	Summer in Northern Hemisphere; low pressure over Asia pulls air northward.	Summer in Southern Hemisphere; low pressure moves south.
ITCZ Position	ITCZ shifts north of Sri Lanka.	ITCZ shifts south of Sri Lanka.
Wind & Origin	Winds enter from Southwest after long passage across South Indian Ocean (high moisture content).	Winds enter from Northeast, passing largely over North Indian landmass (lower moisture content).
Rainfall Intensity	Higher total rainfall and higher intensity.	Relatively lower overall rainfall volume and intensity.

Cropping Seasons in Sri Lanka

- **Yala Season:** March to September (Includes FIM & SWM). Rain is majorly restricted to the Wet Zone; Dry Zone requires irrigation.
- **Maha Season:** October to February (Includes SIM & NEM). Widespread rain across the island, providing crucial water for Dry Zone agriculture.

Effects of Excess / Adverse Rainfall on Crops

- Reduced solar radiation leads to decreased photosynthesis and poor growth.
- Crop lodging (falling over of plants/cereals).
- Shedding and washing away of pollen and flowers.
- Fruit rotting and pre-harvest seed germination.
- Plant death due to flooding and root asphyxiation.
- Leaf tearing and mechanical damage, diminishing photosynthetic surface area.

Detailed Effects of Light on Crops

1. Light Intensity

- **Photosynthesis:** Increased light intensity up to saturation point increases photosynthetic rate.
- **Pigment Synthesis:** Essential for synthesis of chlorophyll, anthocyanins, etc.
- **Respiration & Transpiration:** Higher light intensity opens stomata, increasing transpiration and respiration rates.
- **Phototropism:** Directional stem curvature toward light sources due to auxin distribution.

2. Light Quality (Wavelengths)

- **Red Light:** Promotes branching and seed germination.
- **Far-Red Light:** Inhibits seed germination and promotes stem elongation.
- **Blue-Violet Light:** Enhances internode extension and hypocotyl elongation.
- **Green Light:** Facilitates leaf cell expansion.
- **Yellow Light:** Affects internode elongation and seed setting.
- **Blue & Red Light:** Primary spectral wavelengths absorbed for photosynthesis.

3. Photoperiodism (Light Duration)

Response of plants to relative lengths of daylight and darkness, regulated by the pigment **Phytochrome**.

Category	Photoperiodic Requirement	Crop Examples
Short-Day Plants (SDP)	Require short days / long uninterrupted nights to trigger flowering.	Rice, Coffee, Tobacco, Strawberry, Soybean, Winged Bean, Sesame
Long-Day Plants (LDP)	Require long days / short nights to trigger flowering.	Beetroot, Carrot, Lettuce, Radish, Potato, Spinach, Onion
Day-Neutral Plants (DNP)	Flowering is insensitive to day length duration.	Coconut, Tea, Chilli, Cowpea, Mungbean, French Bean, Bitter Gourd, Snake Gourd, Cucumber

Detailed Effects of Wind on Crops

Favorable Effects

- Gentle breeze refreshes CO₂ around canopy, enhancing photosynthesis.
- Facilitates wind pollination (anemophily) in grasses and grains.
- Essential for monsoonal rainfall transport.
- Harnessed via windmills for pumping underground water.
- Traditional winnowing cleans chaff/impurities from harvested grain.

Adverse Effects

- Excessive evapotranspiration leads to rapid moisture stress.
- Severe lodging and stem breakage in cereal crops.
- Tearing of leaves damages photosynthetic capacity.
- Facilitates spread of airborne fungal spores, pests, and diseases.
- Causes flower and premature fruit drop.

Effects of Temperature on Crops

Factors causing temperature variation include altitude, geographic location, vegetation density, human activity, proximity to water bodies, and rainfall distribution.

- **Impacts:** Governs seed germination rates, photosynthetic enzymatic activity, respiration rates, hormone regulation, flowering induction (vernalization), and tuber formation.

- **Low Temperature Impacts:** Stunted growth, delayed germination, reduced metabolic activity, and cold injury.
- **High Temperature Impacts:** Heat stress, enzyme denaturation, excessive transpiration, pollen sterility, and rapid flower/fruit abortion.

Effects of Relative Humidity (RH) on Crops

Favorable Effects	Adverse Effects
Optimum RH maintains ideal transpiration rates without severe moisture stress.	High RH fosters fungal/bacterial disease outbreaks (e.g., Tea Blister Blight).
High RH aids root initiation in vegetative stem cuttings.	Encourages pest proliferation (e.g., Brown Plant Hopper, Whiteflies).
Prolongs stigma receptivity in flowers during pollination.	Inhibits wind-borne pollen dispersion due to clumping.
Enhances flower quality and vase life in cut flowers (e.g., Anthurium, Orchids).	Reduces transpiration rate, impeding nutrient transport (calcium/boron).
Regulates stomatal opening and closing dynamics.	High humidity complicates seed storage, increasing spoilage and fungal rot.

Agro-Meteorological Station Setup & Operation

An Agro-Meteorological Station collects standardized weather parameters vital for agricultural planning.

Site Selection Criteria

- Located on land representative of the broader agricultural region.
- Free from nearby large water bodies, ponds, or artificial reservoirs.
- Flat, well-drained level ground covered with short, frequently trimmed grass.
- Distance from obstructions (trees/buildings) must be at least **4 times the height** of the obstruction.
- Easily accessible with clear pathways.
- Standard plot dimensions: **10 m × 10 m** core plot within a **50 m × 50 m** surrounding open area, secured with a wire fence.

Standard Instrument Installation Rules

Instrument	Installation Guidelines
Simple Rain Gauge	Installed on a concrete platform so the top rim sits exactly 30 cm above ground level.
Anemometer & Wind Vane	Mounted on a standard pole 2.0 meters above ground level.

Instrument	Installation Guidelines
Class A Evaporation Pan	Placed on a wooden pallet 15 cm off the ground, situated 1.5 m inside the fence perimeter and at least 5 m away from the rain gauge.
Thermometers (Max/Min/ Psychrometer)	Mounted on a wooden panel inside a Stevenson Screen at eye level (1.2 m to 1.5 m height).
Soil Thermometers	Inserted directly into soil at depths of 5 cm, 10 cm, 20 cm, 30 cm, and 100 cm .
Sunshine Recorder	Mounted on a concrete pillar (30 cm × 30 cm surface) at 1.5 m height, strictly aligned along the East-West axis.

Data Collection Schedule

- Standard Observation Hours: **08:30 hrs (8:30 AM)** and **15:30 hrs (3:30 PM)**.
- **Once Daily (at 08:30 hrs):** Rainfall, Max/Min Temperatures, Sunshine Hours, Wind Speed, Evaporation. Recorded under the previous date.
- **Twice Daily (at 08:30 & 15:30 hrs):** Dry/Wet Bulb Temperatures and Soil Temperatures. Recorded under current date (Morning / Afternoon).

Climate Change & Its Impact on Agriculture

Climate Change refers to long-term shifts in global or regional climate patterns, driven significantly by global warming.

Natural vs. Accelerated Greenhouse Effect

- **Natural Greenhouse Effect:** Solar radiation warms Earth's surface; re-radiated infrared rays are absorbed by atmospheric greenhouse gases, maintaining habitability.
- **Accelerated Greenhouse Effect:** Human industrial activities increase concentrations of greenhouse gases, trapping excessive heat and accelerating global warming.

Key Greenhouse Gases

- Carbon Dioxide (CO_2)
- Methane (CH_4)
- Nitrous Oxide (N_2O)
- Chlorofluorocarbons (CFCs)
- Sulfur Hexafluoride (SF_6)

Impacts of Climate Change on Agriculture

- **Rainfall Variability:** Prolonged droughts, unseasonal heavy rains, high-intensity flash downpours.
- **Temperature Rises:** Shifts in minimum/maximum temperature ranges (e.g., recorded rises in Mahailuppallama, Anuradhapura, and Nuwara Eliya).

- **Agricultural Effects:** Crop yield reduction, proliferation of invasive weeds, water scarcity, El Niño/La Niña oscillations, increased pest/disease outbreaks, shortened crop growth cycles, and sea-level rise causing coastal soil salinization.
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| Climatic, Agro-Climatic & Agro-Ecological Zones of Sri Lanka

1. Main Climatic Zones (Based on Annual Rainfall)

- **Wet Zone:** Annual rainfall > 2500 mm
- **Intermediate Zone:** Annual rainfall between 1750 mm – 2500 mm
- **Dry Zone:** Annual rainfall < 1750 mm

2. Agro-Climatic Zones (9 Sub-divisions)

Divided based on elevation (Low-country <300m, Mid-country 300-1000m, Up-country >1000m) across the 3 main climatic zones:

- Low-country Wet Zone, Mid-country Wet Zone, Up-country Wet Zone
- Low-country Intermediate Zone, Mid-country Intermediate Zone, Up-country Intermediate Zone
- Low-country Dry Zone

3. Agro-Ecological Zones (AEZs)

An **Agro-Ecological Region** is a land unit uniform in climate, soil profile, landform, and land-use suitability.

- Sri Lanka is divided into **46 Agro-Ecological Zones** (2003 classification).
- **Distribution:** 15 in the Wet Zone, 20 in the Intermediate Zone, and 11 in the Dry Zone.

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